INCH-POUND

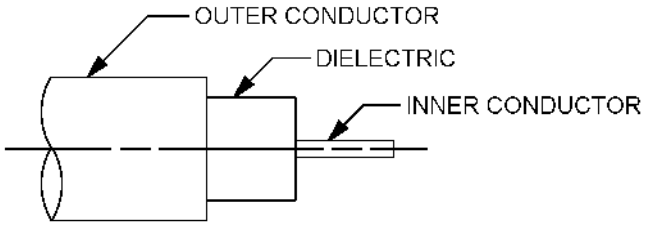
MIL-DTL-17/129F
09 May 2019
SUPERSEDING
MIL-DTL-17/129E
26 July 2006

DETAIL SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, COAXIAL, 0.250 (6.35 mm) DIAMETER, SEMIRIGID, 50 OHMS, M17/129-RG401 AND M17/129-00001

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17.



Inches	mm
.001	0.03
.002	0.05
.0641	1.682
.209	5.31
.250	6.35

* Metric conversions for table I

FIGURE 1. Configuration.

TABLE I. Description.

Part or Identifying Number (PIN)	Inner conductor	Dielectric core	Outer conductor 1/
M17/129-RG401	Solid, silver-coated copper wire diameter, .0641 inch ± .0010	Type F-1 Diameter, .209 inch ± .002	Copper tubing diameter, .250 inch ± .001
M17/129-00001	Same as above	Same as above	Copper tubing diameter, .250 inch ± .002, -.001 Tin plated in accordance with ASTM B545, .0003 minimum inch thick 2/

1/ Welded outer conductor in accordance with ASTM B447 and MIL-DTL-17 is optional.
2/ The outer diameter dimensions is after plating.

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Engineering information:

Continuous working voltage: 3,000 V rms, maximum.

Operating frequency: 18 GHz, maximum.

Velocity of propagation: 69.5 percent, nominal.

Power rating: See figure 2.

Operating temperature range: -40°C to + 90°C.

Inner conductor properties:

DC resistance (maximum at 20°C): .257 ohms per 100 feet.

Elongation: 25 percent, minimum, per 10-inch specimen.

Tensile strength: Applicable.

Engineering notes: This cable is generally useful in critical RF performance applications (see connector series TNC in accordance with MIL-PRF-39012). This cable is generally manufactured in 20-foot lengths. Different lengths are available.

REQUIREMENTS:

Dimensions, configuration and description: See figure 1 and table I.

Environmental and mechanical:

Visual and mechanical examination: Applicable.

Diameter: Applicable.

Out-of-roundness: Applicable.

Eccentricity: 7.5 percent, maximum.

Adhesion of conductors:

Inner conductor to core: 4 pounds minimum; 100 pounds, maximum.

Outer conductor to core: 4 pounds minimum.

Electrical and mechanical:

Test frequency: 400 MHz to 18 GHz.

Continuity: Applicable.

Spark test: Not Applicable.

Voltage withstanding: 7,500 V rms, minimum

Insulation resistance: Not applicable.

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Corona extinction voltage: 3,000 V rms, minimum.

Characteristic impedance: 50 ± .5 ohms.

RF transmission loss (attenuation): See figure 2.

Standing wave ratio (return loss): See figure 3. Minimum cable length 7 feet.

Capacitance: 29.6 pF per foot, maximum.

Capacitance stability:	Not applicable.
Capacitance unbalance:	Not applicable.
Transmission unbalance:	Not applicable.
Inductance:	Not applicable.
Mechanically induced noise:	Not applicable.
Time delay:	Not applicable.
Aging stability:	Not applicable.
Stress crack resistance:	Not applicable.

Outer conductor integrity: Applicable

Cold bend: Not applicable.

Dimensional stability: +125°C ± 5°C.

Core from outer conductor: .015 inch, maximum.

Contamination: Not applicable.

Bendability: Mandrel diameter .75 inch, maximum.

Flammability: Not applicable..

Flame propagation:	Not applicable.
Acid gas generation:	Not applicable.
Halogen:	Not applicable.
Fluid immersion:	Not applicable.
Smoke index:	Not applicable.
Toxicity index:	Not applicable.
Durometer hardness:	Not applicable.
Weathering:	Not applicable.
Abrasion resistance:	Not applicable.
Tear strength:	Not applicable.
Heat distortion:	Not applicable.
Physicals:	Not applicable.

Tensile strength and elongation: Not applicable.

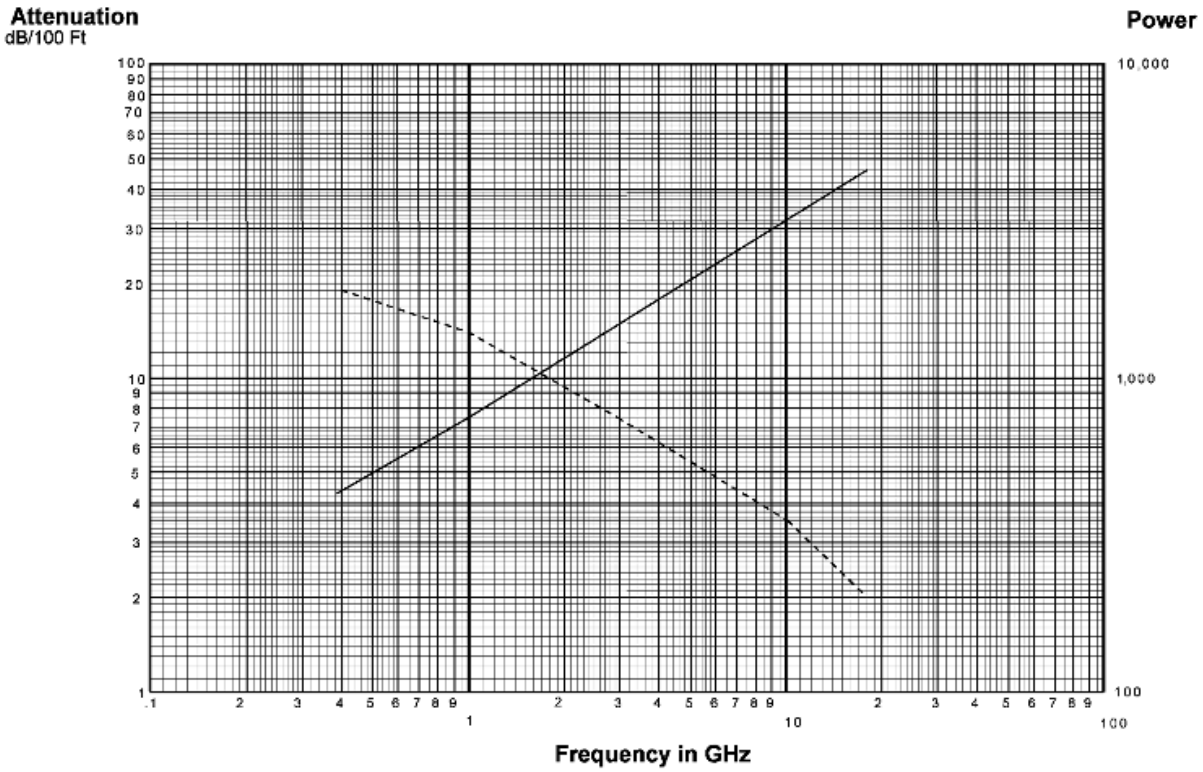
Weight: 10.5 pounds per 100 feet, maximum (M17/129-RG401).
10.6 pounds per 100 feet, maximum (M17/129-00001).

Marking: Not Applicable.

Workmanship: Applicable.

PIN: See table I.

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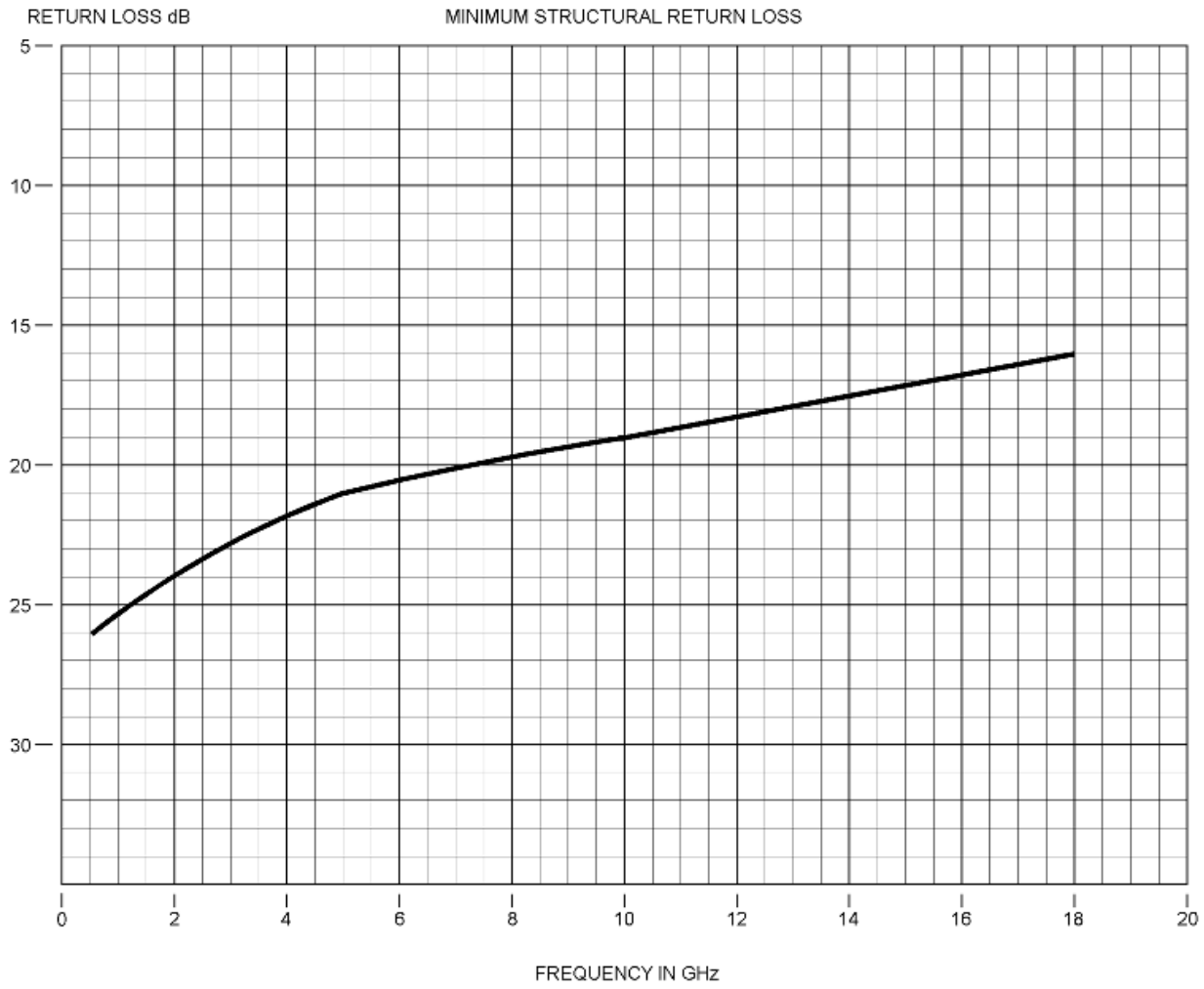
Maximum attenuation: ———————— (Test requirements shall be a line indicted on graph)

Maximum power - - - - - - - - - - - - - - - - -
(at 25°C at sea level)

Attenuation		Power
MHz	dB	Watts
400	4.5	1900
1000	7.5	1400
3000	16	750
10000	33	350
18000	48	200

FIGURE 2. Power rating and attenuation.

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Structural Return Loss	
MHz	dB
500	26
5000	21
10000	19
18000	16

FIGURE 3. Structural return loss.

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SWR	REFLECTION COEFFICIENT	RETURN LOSS dB	SWR	REFLECTION COEFFICIENT	RETURN LOSS dB
17.3910	.8913	1	1.3767	.1585	16
8.7242	.7943	2	1.3290	.1413	17
5.8480	.7079	3	1.2880	.1259	18
4.4194	.6310	4	1.2528	.1122	19
3.5698	.5623	5	1.2222	.1000	20
3.0095	.5012	6	1.1957	.0891	21
2.6146	.4467	7	1.1726	.0794	22
2.3229	.3981	8	1.1524	.0708	23
2.0999	.3548	9	1.1347	.0631	24
1.9250	.3162	10	1.1192	.0562	25
1.7849	.2818	11	1.1055	.0501	26
1.6709	.2512	12	1.0935	.0447	27
1.5769	.2239	13	1.0829	.0398	28
1.4985	.1995	14	1.0736	.0355	29
1.4326	.1778	15	1.0653	.0316	30

FIGURE 3. Structural return loss - Continued.

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TABLE II. Qualification and Group A, B, and C Table.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
I	In-process inspection:		4.5	Entire sample		
	Continuity	3.7.1	4.8.2	Entire sample		
	Spark test	N/A				
	Voltage withstanding	3.7.3	4.8.4	Entire sample		
	Insulation resistance	N/A				
	Visual and mechanical inspection	3.6	4.8.1	Entire sample	Per Table VIII	A
	Diameter measurements.	3.6.1	4.8.1.1	Entire sample	Per Table VIII	A
	Out-of-roundness	3.6.2	4.8.1.2	Entire sample	Per Table VIII	A
	Eccentricity of inner conductor	3.6.3	4.8.1.3	2	Per Table VIII	A
	Adhesion of conductors	3.6.4	4.8.1.4	4	Per Table VIII	A
	Marking	N/A				
	Workmanship	3.10	4.8.1	Entire sample	Per Table VIII	A
II	Corona extinction voltage	3.7.5	4.8.6	1	Per Table VIII	B
	Characteristic impedance	3.7.6	4.8.7	1	Per Table VIII	A
	RF transmission loss (attenuation)	3.7.7	4.8.8	2	Per Table VIII	A
	Standing wave ratio (return loss)	3.7.8	4.8.9	2	Per Table VIII	A
	Capacitance	3.7.9	4.8.10	1	Per Table VIII	B
	Capacitance stability	N/A				
	Capacitance unbalance	N/A				
	Transmission unbalance	N/A				
	Inductance	N/A				
	Mechanically induced noise voltage	N/A				
	Time delay	N/A				
	Aging stability	N/A				
	Stress crack resistance	N/A				
	Outer conductor integrity	3.7.17	4.8.18	4	Per Table VIII	C

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TABLE II. Qualification and Group A, B, and C Table - Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Qual- # of Specimens	Conf - # of Specimens	Conformance inspection
II	Cold bend	N/A				
	Dimensional stability	3.7.19	4.8.20	1	Per Table VIII	C
	Contamination	N/A				
	Bendability	3.7.21	4.8.22	2	Per Table VIII	C
	Flammability	N/A				
	Flame propagation	N/A				
	Acid gas generation	N/A				
	Halogen content	N/A				
	Immersion tests	N/A				
	Smoke index	N/A				
	Toxicity Index	N/A				
	Durometer hardness	N/A				
	Weathering	N/A				
	Abrasion resistance	N/A				
	Tear strength	N/A				
	Heat distortion	N/A				
	Physicals (aged)	N/A				
	Tensile strength and elongation	N/A				
	Weight	3.9	4.8.24	1	Per Table VIII	B

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Supersession data: See table III.

TABLE III. Cross reference of PIN.

PIN	Superseded PIN or type designation
M17/129-RG401	RG-401/U
M17/129-00001	--

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-DTL-17, this document references the following:

ASTM B447
ASTM B545
MIL-PRF-39012

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2018-060)

Review activities:
Army - AT, AV, CR4, MI
Navy - AS, MC, OS, SH
Air Force - 19
DLA - CQ

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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